

Optical properties of thin films up to second order in the thickness

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Abstract

In this paper we give expressions for the reflection and transmission amplitudes of isotropic non-magnetic films for which the average thickness and the correlation length, which describe the film structure, are small in comparison with the wavelength of the incident light. It is shown that these expressions can be expressed consistently up to second order in the ratio of the thickness to the wavelength in four universal constitutive coefficients. These four coefficients are found to be related to the density of the static dipole and quadrupole polarizabilities in the film. For island films on a plane substrate and for stratified media, we give explicit expressions for these coefficients and discuss our results by comparing with experimental results and with other theoretical approaches. As an example we calculate numerically the optical properties of gold spheres on a sapphire substrate as a function of the coverage and the frequency. A quadrupole approximation is introduced, which is found to be accurate within a few percent up to coverages of about 50% compared with exact numerical results which use multipoles to arbitrary order.

Keywords: Dielectric properties; Ellipsometry; Gold; Optical properties

1. Introduction

In this article we give a general description of isotropic non-magnetic films for which the average thickness (or average particle size for island films) and the correlation length (or pair correlations between the particles), which describes the film structure, are small in comparison with the wavelength of the incident light. In Section 2.1 we introduce the general description used to calculate the optical properties of such films. The basic idea is to replace the film by a dividing surface directly separating the two dielectric media. At this dividing surface, one locates excess surface polarization and magnetization densities chosen in such a way that the electromagnetic fields outside the original film remain the same. These excess densities then result in a modification of the usual Fresnel boundary conditions at the dividing surface between two dielectric media. These modifications are described in terms of two constitutive coefficients, β and γ , which are of first order in the film thickness, and two constitutive coefficients, δ and τ are of second order in the film thickness. These constitutive coefficients are found to depend on the choice of location of the dividing surface. Explicit expressions are given for the dependence of the constitu-

tive coefficients on this choice. It is obvious that the optical properties are independent of this choice, since the excess densities are chosen in such a manner that the electromagnetic fields outside the film remain the same. As a consequence one can define three optical invariants which do not depend on the choice of the dividing surface. These invariants are combinations of the different constitutive coefficients. If the constitutive coefficients are complex, the three invariants are also complex. In that case one may define one more real invariant. In Section 2.2 we give the reflection and transmission amplitudes in terms of the constitutive coefficients. The resulting reflectance, transmittance, adsorption and ellipsometric coefficients are given in terms of the optical invariants.

In Section 3.1 the preceding theory is applied to island films on a plane substrate and in section 3.2 to stratified media. It is shown that our theory gives the same results as found using conventional methods [7] for stratified media. The dependence of the constitutive coefficients on the choice of the position of the dividing surface is used to derive expressions for these coefficients in terms of dipole and quadrupole polarizabilities. For both cases it is found that the constitutive coefficients β and γ are related to the density of the

static dipole polarizabilities in the film and that δ and τ are related to the density of the static quadrupole polarizabilities in the film. Retardation effects are shown to be of third and higher order and are therefore not important for the calculation of these coefficients. In Section 4 we compare the quadrupole model with the results of exact calculations up to arbitrary order in the multipole expansion for a regular array of spherical gold islands. It is concluded that up to a coverage of 50% the quadrupole model is correct within a few percent. In the last section we summarize and briefly describe the application of the quadrupole model to a system of latex spheres adsorbed on a glass substrate. Available experimental results [6] found to be explained by the theory, without adjustable parameters.

In a previous article [1] we gave expressions to calculate the reflection and transmission amplitudes of films consisting of small monodisperse metal spheres. In this paper we performed an exact calculation of the induced multipoles in the spheres, in which the interaction with other spheres and with the image charge distribution in the substrate were taken into account. In the reflection and transmission amplitudes we neglected, however, the second order constitutive coefficients δ and τ . Calculations were done in order to investigate whether the second order contributions lead to adjustments of the conclusions formulated in our first paper [1]. It is found that the corrections, though usually small, are nevertheless large enough to be measurable with the available accuracy. For non-absorbing particles, these corrections are in general more important. The conclusions formulated in our first paper, however, remain valid. In this previous article matrix equations were derived for the multipole polarizabilities of spheres on a substrate in an external homogeneous electric field $\vec{E}_0$. For one sphere with radius R one finds, in terms of these matrix elements for the dipole polarization:

$$\left. \begin{aligned} p_{\parallel} &= M_{11}^{11} E_{0,\parallel} \\ p_{\perp} &= M_{10}^{10} E_{0,\perp} \end{aligned} \right\} \times 4\pi\epsilon_a R^3 \frac{\epsilon - \epsilon_a}{\epsilon + 2\epsilon_a} \quad (1.1)$$

and for the quadrupole polarization:

$$\left. \begin{aligned} q_{\perp,\parallel} &= q_{\parallel,\perp} = \sqrt{5} M_{21}^{11} E_{0,\parallel} \\ q_{\perp,\perp} &= -2q_{\parallel,\parallel} = 2\sqrt{5/3} M_{20}^{10} E_{0,\perp} \end{aligned} \right\} \times 4\pi\epsilon_a R^4 \frac{\epsilon - \epsilon_a}{\epsilon + 2\epsilon_a} \quad (1.2)$$

The dipole as well as the quadrupole polarization are defined with respect to the center of the sphere. The subscript $\parallel$ means parallel to the ambient/substrate interface and the subscript $\perp$ means normal to this interface. The dielectric constants of the sphere, ambient and substrate are given by ϵ , ϵ_a and ϵ_s , respectively.

In Section 4 we present the results of the evaluation of the dielectric and optical properties of triangular arrays of gold spheres with a radius of 15 nm on a sapphire substrate in vacuum as a function of the

coverage and the wavelength. The behaviour of the random distribution and the square array, as discussed in the previous article, are rather similar and would not add much to the analysis in this paper. In Section 4.1 the quadrupole approximation is introduced, which is found to give accurate results up to a coverage of about 50%. In Section 4.2 we study the ellipsometric properties of these films. In particular the ellipsometric angles Δ and ψ at the Brewster angle of the bare substrate are studied, along with the shift of the Brewster angle owing to the film material. In Section 5 we summarize our results and formulate conclusions.

2. Theory

2.1. Description of thin non-magnetic isotropic films in terms of constitutive coefficients and their dependence on the choice of the dividing surface

For the calculation of the reflection and transmission amplitudes of a finite boundary layer between dielectric media it is convenient to extrapolate the electromagnetic fields to a flat mathematical interface. In this so-called dividing surface, situated somewhere within the original layer and separating directly the two media, one locates fluctuating excess surface polarization, $\vec{P}^s$, and magnetization, $\vec{M}^s$, densities such that the electromagnetic fields remain the same outside the original film. The boundary conditions at the dividing surface for the parallel component of the electric field, the normal component of the displacement field, the parallel component of the magnetic field and the normal component of the magnetic induction, in terms of these surface excess densities, are respectively given by [2,3]:

$$\vec{E}_{\parallel,-}(\vec{r}_{\parallel}, 0, \omega) = -\vec{\nabla}_{\parallel} P_{\perp}^s(\vec{r}_{\parallel}, 0, \omega) - i \frac{\omega}{c} \hat{z} \times \vec{M}_{\parallel}^s(\vec{r}_{\parallel}, 0, \omega) \quad (2.1)$$

$$D_{\perp,-}(\vec{r}_{\parallel}, 0, \omega) = -\vec{\nabla}_{\parallel} \cdot \vec{P}_{\parallel}^s(\vec{r}_{\parallel}, 0, \omega) \quad (2.2)$$

$$\vec{H}_{\parallel,-}(\vec{r}_{\parallel}, 0, \omega) = -\vec{\nabla}_{\parallel} M_{\perp}^s(\vec{r}_{\parallel}, 0, \omega) + i \frac{\omega}{c} \hat{z} \times \vec{P}_{\parallel}^s(\vec{r}_{\parallel}, 0, \omega) \quad (2.3)$$

$$B_{\perp,-}(\vec{r}_{\parallel}, 0, \omega) = -\vec{\nabla}_{\parallel} \cdot \vec{M}_{\parallel}^s(\vec{r}_{\parallel}, 0, \omega) \quad (2.4)$$

Here the subscript $-$ indicates the jump in the extrapolated fields at the dividing surface, located at $z=0$, going from $z < 0$ to $z > 0$. The z axis of the coordinate system is chosen normal to the ambient/substrate interface in the direction of the incident light. The symbol $\hat{z}$ denotes the unit vector in the z direction. The speed of light in the vacuum is denoted by c while ω is equal to 2π times the frequency of the incident light.

In this paper we consider isotropic films for which the averaged thickness and the (correlation) length

which describes the film structure are small in comparison with the wavelength of the incident light. As a result the film can, to lowest order in the film thickness, be described by two susceptibilities which give the excess surface polarization density:

$$\bar{P}_{\parallel}^s(\vec{r}_{\parallel}, 0, \omega) = \gamma(0)\bar{E}_{\parallel,+}(\vec{r}_{\parallel}, 0, \omega) \quad (2.5)$$

$$P_{\perp}^s(\vec{r}_{\parallel}, 0, \omega) = \beta(0)D_{\perp,+}(\vec{r}_{\parallel}, 0, \omega) \quad (2.6)$$

$$\vec{M}_{\parallel}^s(\vec{r}_{\parallel}, 0, \omega) = 0 \quad (2.7)$$

$$M_{\perp}^s(\vec{r}_{\parallel}, 0, \omega) = 0 \quad (2.8)$$

in which the subscript + indicates half the sum of the extrapolated fields on both sides of the dividing surface. The excess surface magnetization density is zero to lowest order. Both γ and β are of first order in the film thickness.

In our previous article [1] for spheres on a substrate, we found that the constitutive coefficients γ and β are related to the average polarizability per unit surface area and we placed the dividing surface as well as the $X-Y$ plane on the ambient/substrate interface. It is obvious that the phase difference between light reflected from the spheres and light reflected from the ambient/substrate interface has been neglected in the above constitutive equations. Physically this means that we have not taken into account the effect of shifting the polarization from the centers of the spheres to the substrate. In other words, we have taken into account the total polarization of the film but not the influence of the distribution of the polarization inside the film.

In order to find the influence of this phase factor and thereby of the polarization distribution inside the film, we shift the dividing surface from $z = 0$ to d with $d > 0$ and change the excess surface densities in such a manner that the resulting electromagnetic fields for $z > d$ and $z < 0$ remain unchanged. The jump in the dielectric constant now occurs at $z = d$ where it changes from ϵ_a for $z < d$ to ϵ_s for $z > d$. Consider for instance the discontinuity at the dividing surface at $z = 0$ of the x -component of the electric field

$$E_x(+\eta) - E_x(-\eta) = -\frac{1}{2}\beta(0)\partial_x\{D_z(+\eta) + D_z(-\eta)\} \quad (2.9)$$

For convenience the variables $\vec{r}_{\parallel}$ and ω have been omitted; η is a positive infinitesimally small number. Up to second order one has

$$E_x(+\eta) = E_x(d+\eta) - d\partial_z E_x(z)|_{z=d+\eta} + \frac{1}{2}d^2(\partial_z)^2 E_x(z)|_{z=d+\eta} \quad (2.10)$$

Using $\vec{\nabla} \times \vec{E} = i\omega/c\vec{B}$ and $\Delta\vec{E} + (\omega^2/c^2)\epsilon_s\vec{E} = 0$, one can eliminate the normal derivatives

$$E_x(+\eta) = E_x(d+\eta) - d\{\partial_x E_x(d+\eta) + i\frac{\omega}{c}B_y(d+\eta)\} - \frac{1}{2}d^2\left\{\frac{\omega^2}{c^2}\epsilon_s + \Delta_{\parallel}\vec{E}_{\parallel}(d+\eta)\right\} \quad (2.11)$$

Since at the right hand side of Eq. (2.9) β is of first order in the film thickness, to first order one only needs

$$\partial_x D_x(+\eta) = \partial_x\{D_z(d+\eta) - d\partial_x D_z(z)|_{z=d+\eta}\} \quad (2.12)$$

Using $\vec{\nabla}\vec{D} = 0$, one can also eliminate the normal derivative here:

$$\partial_x D_z(+\eta) = \partial_x\{D_z(d+\eta) + d\vec{\nabla}_{\parallel} \cdot \vec{D}_{\parallel}(d+\eta)\} \quad (2.13)$$

Proceeding in this manner for the other fields as well, one can extrapolate the electromagnetic fields from 0 to d and find the boundary conditions for a dividing surface located at $z = d$ but with the same electromagnetic properties as for the dividing surface located at $z = 0$. For the excess surface densities one finds, after some elaborate algebra up to second order in the film thickness,

$$\begin{aligned} \bar{P}_{\parallel}^s(\vec{r}_{\parallel}, d, \omega) &= \gamma(d)\bar{E}_{\parallel,+}(\vec{r}_{\parallel}, d, \omega) \\ &\quad - \delta(d)\vec{\nabla}_{\parallel} D_{\perp,+}(\vec{r}_{\parallel}, d, \omega) \\ &\quad + i\frac{\omega}{c}\tau(d)_z \times \vec{H}_{\parallel,+}(\vec{r}_{\parallel}, d, \omega) \end{aligned} \quad (2.14)$$

$$\begin{aligned} \bar{P}_{\perp}^s(\vec{r}_{\parallel}, d, \omega) &= \beta(d)D_{\perp,+}(\vec{r}_{\parallel}, d, \omega) \\ &\quad + \delta(d)\vec{\nabla}_{\parallel} \cdot \vec{E}_{\parallel,+}(\vec{r}_{\parallel}, d, \omega) \end{aligned} \quad (2.15)$$

$$\vec{M}_{\parallel}^s(\vec{r}_{\parallel}, d, \omega) = i\frac{\omega}{c}\tau(d)_z \times \vec{E}_{\parallel,+}(\vec{r}_{\parallel}, \omega) \quad (2.16)$$

$$M_{\perp}^s(\vec{r}_{\parallel}, d, \omega) = 0 \quad (2.17)$$

Here we have defined two constitutive coefficients δ and τ which are of second order in the film thickness. The excess polarization and magnetization densities are now related to the fields in a non-local manner, to this order in the thickness over the wavelength they depend on the linear spacial and temporal gradients of the fields. The reason for this is that the shifted excess polarization density is coupled to the fields above and below the original dividing surface. This naturally leads to a non-local dependence on the fields.

For this dividing surface the constitutive coefficients γ , β , δ and τ are given by

$$\begin{aligned} \gamma(d) &= \gamma(0) + (\epsilon_s + \epsilon_a)d \\ \beta(d) &= \beta(0) - \left(\frac{1}{\epsilon_s} - \frac{1}{\epsilon_a}\right)d \\ \tau(d) &= \tau(0) + \gamma(0)d + \frac{1}{2}(\epsilon_s - \epsilon_a)d^2 \\ \delta(d) &= \delta(0) + \frac{1}{2}\gamma(0)d(1/\epsilon_a + 1/\epsilon_s) + \frac{1}{2}\beta(0)d(\epsilon_a + \epsilon_s) \\ &\quad + \frac{1}{2}(\epsilon_s + \epsilon_a)\left(\frac{1}{\epsilon_s} - \frac{1}{\epsilon_a}\right)d^2 \end{aligned} \quad (2.18)$$

Notice that we have added $\delta(0)$ and $\tau(0)$ since in general one cannot choose the location of the dividing surface in such a manner that both coefficients are equal to zero. The constitutive coefficients depend on the choice of the dividing surface. The electromagnetic fields far from the region $0 \leq z \leq d$ are invariant with respect to this choice. The optical properties of the film must therefore depend on combinations of the constitutive coefficients which are invariant with respect to the choice of the dividing surface. One can define three complex invariants

$$\begin{aligned} I_e &\equiv \gamma - \varepsilon_a \varepsilon_s \beta \\ I_\tau &\equiv \tau - \frac{1}{2(\varepsilon_s - \varepsilon_a)} (\gamma)^2 \\ I_{\delta e} &\equiv \delta - \frac{\varepsilon_s + \varepsilon_a}{2(\varepsilon_s - \varepsilon_a)} \gamma \beta \end{aligned} \quad (2.19)$$

and one real invariant

$$I_c \equiv (\varepsilon_s - \varepsilon_a)'' \gamma' - (\varepsilon_s - \varepsilon_a)' \gamma'' \quad (2.20)$$

The prime and double prime indicate the real and imaginary part of the corresponding quantity, respectively. Notice the fact that if both ε_s and ε_a are non-absorbing that both γ'' and β'' are invariant. In that case it is convenient to use these quantities rather than I''_e and I_c .

If one has two dividing surfaces located at z_1 and z_2 , with $z_2 - z_1 = \eta$ where η is a positive infinitesimally small number, then it is possible to combine these two dividing surfaces into one by eliminating the electromagnetic fields between z_1 and z_2 . The constitutive coefficients of this new dividing surface are found to be equal to

$$\begin{aligned} \gamma_{\text{tot}} &= \gamma_1 + \gamma_2 \\ \beta_{\text{tot}} &= \beta_1 + \beta_2 \\ \tau_{\text{tot}} &= \tau_1 + \tau_2 \\ \delta_{\text{tot}} &= \delta_1 + \delta_2 - \frac{1}{2} \gamma_1 \beta_2 + \frac{1}{2} \gamma_2 \beta_1 \end{aligned} \quad (2.21)$$

These addition formulae will be used in Section 3.2 to calculate the constitutive coefficients for a stratified layer.

In the analysis of the rest of the paper, we will usually use the value of the constitutive coefficients on the surface of the substrate. For ease of notation we will then suppress the explicit dependence on the position and then use γ , β , δ and τ for these values.

2.2. The reflectance, the transmittance, the adsorption and the ellipsometric coefficient up to second order in the film thickness

The reflection and transmission amplitudes can be calculated using the Maxwell equations and the

boundary conditions, Eqs. (2.1)–(2.4) with Eqs. (2.14)–(2.17) for the dividing surface [2,3]. For light incident from the ambient, we find in this way

$$\begin{aligned} r_s &= \left\{ n_a \cos \theta \left[1 + \left(\frac{\omega}{c} \right)^2 \tau \right] \right. \\ &\quad \left. - n_s \cos \theta_t \left[1 - \left(\frac{\omega}{c} \right)^2 \tau \right] + i(\omega/c)\gamma \right\} / d_s(\theta) \end{aligned} \quad (2.22)$$

$$t_s = 2n_a \cos \theta / d_s(\theta) \quad (2.23)$$

for s-polarized light with the denominator given by

$$\begin{aligned} d_s(\theta) &\equiv \left\{ n_a \cos \theta \left[1 + \left(\frac{\omega}{c} \right)^2 \tau \right] \right. \\ &\quad \left. + n_s \cos \theta_t \left[1 - \left(\frac{\omega}{c} \right)^2 \tau \right] - i(\omega/c)\gamma \right\} \end{aligned}$$

and for p-polarized light

$$\begin{aligned} r_p &= \left\{ n_s \cos \theta \left[1 - \left(\frac{\omega}{c} \right)^2 \tau(\theta) \right] \right. \\ &\quad \left. - n_a \cos \theta_t \left[1 - \left(\frac{\omega}{c} \right)^2 \tau(\theta) \right] \right. \\ &\quad \left. - i \frac{\omega}{c} [\gamma \cos \theta \cos \theta_t - n_a n_s \beta \varepsilon_a \sin^2 \theta] \right. \\ &\quad \left. - \frac{1}{4} \left(\frac{\omega}{c} \right)^2 \gamma \beta \varepsilon_a \sin^2 \theta (n_s \cos \theta - n_a \cos \theta_t) \right\} / d_p(\theta) \end{aligned} \quad (2.24)$$

$$t_p = 2n_a \cos \theta \left[1 + \frac{1}{4} \left(\frac{\omega}{c} \right)^2 \gamma \beta \varepsilon_a \sin^2 \theta \right] / d_p(\theta) \quad (2.25)$$

with the denominator given by

$$\begin{aligned} d_p(\theta) &\equiv n_s \cos \theta \left[1 - \left(\frac{\omega}{c} \right)^2 \tau(\theta) \right] \\ &\quad + n_a \cos \theta_t \left[1 + \left(\frac{\omega}{c} \right)^2 \tau(\theta) \right] \\ &\quad - i \frac{\omega}{c} [\gamma \cos \theta \cos \theta_t + n_a n_s \beta \varepsilon_a \sin^2 \theta] \\ &\quad - \frac{1}{4} \left(\frac{\omega}{c} \right)^2 \gamma \beta \varepsilon_a \sin^2 \theta (n_s \cos \theta + n_a \cos \theta_t) \end{aligned}$$

and the abbreviation

$$\tau(\theta) \equiv \tau - \varepsilon_a \delta \sin^2 \theta$$

The reflectance is given by the absolute square of the reflection amplitude. The transmittance is given by the absolute square of the transmission amplitude times $n_s \cos \theta_t / n_a \cos \theta$. This prefactor originates from the fact that the energy flow orthogonal to the interface is used rather than the energy flow in the direction of propagation. For reflected light this makes no difference. The absorption is equal to one minus the reflectance and the transmittance. Proceeding in this way one finds, after

some tedious algebra, that for films embedded in non-absorbing media

$$R_s(\theta) = \left[\left(n_a \cos \theta - n_s \cos \theta_t - \frac{\omega}{c} \gamma'' \right)^2 + \left(\frac{\omega}{c} \gamma'' \right)^2 - 2 \left(\frac{\omega}{c} \right)^2 (\varepsilon_s - \varepsilon_a) I'_t \right] / D_s(\theta) \quad (2.26)$$

$$T_s(\theta) = 4n_a n_s \cos \theta \cos \theta_t / D_d(\theta) \quad (2.27)$$

$$A_s(\theta) = 4n_a \cos \theta \frac{\omega}{c} \gamma'' / D_s(\theta) \quad (2.28)$$

for s-polarized light with the denominator given by

$$D_s(\theta) = \left[\left(n_a \cos \theta + n_s \cos \theta_t + \frac{\omega}{c} \gamma'' \right)^2 + \left(\frac{\omega}{c} \gamma'' \right)^2 - 2 \left(\frac{\omega}{c} \right)^2 (\varepsilon_s - \varepsilon_a) I'_t \right]$$

and for p-polarized light

$$R_p(\theta) = \left[\left[n_s \cos \theta - n_a \cos \theta_t + \frac{\omega}{c} \times (\gamma'' \cos \theta \cos \theta_t - \varepsilon_a n_a n_s \beta'' \sin^2 \theta) \right]^2 - \left(\frac{\omega}{c} \right)^2 \left\{ 2 \left[I'_t - \frac{1}{2(\varepsilon_s - \varepsilon_a)} (\gamma'')^2 - \varepsilon_a I'_{\delta e} \sin^2 \theta \right] \times [\varepsilon_s \cos^2 \theta - \varepsilon_a \cos^2 \theta_t] - (I'_e)^2 \frac{\varepsilon_a}{\varepsilon_s} \sin^4 \theta + 2\varepsilon_a n_a n_s \gamma'' \beta'' \sin^2 \theta \cos(\theta + \theta_t) \right\} \right] / D_p(\theta) \quad (2.29)$$

$$T_p(\theta) = 4n_a n_s \cos \theta \cos \theta_t / D_p(\theta) \quad (2.30)$$

$$A_p(\theta) = 4n_a \frac{\omega}{c} \cos \theta \left[\gamma'' \cos^2 \theta_t + \varepsilon_a \varepsilon_s \beta'' \sin^2 \theta + 2\varepsilon_a n_s \frac{\omega}{c} \gamma'' \beta'' \cos \theta_t \sin^2 \theta \right] / D_p(\theta) \quad (2.31)$$

with the denominator given by

$$D_p(\theta) = \left[\left[n_s \cos \theta + n_a \cos \theta_t + \frac{\omega}{c} (\gamma'' \cos \theta \cos \theta_t + \varepsilon_a n_a n_s \beta'' \sin^2 \theta) \right]^2 - \left(\frac{\omega}{c} \right)^2 \left\{ 2 \left[I'_t - \frac{1}{2(\varepsilon_s - \varepsilon_a)} (\gamma'')^2 - \varepsilon_a I'_{\delta e} \sin^2 \theta \right] \times [\varepsilon_s \cos^2 \theta - \varepsilon_a \cos^2 \theta_t] - (I'_e)^2 \frac{\varepsilon_a}{\varepsilon_s} \sin^4 \theta - 2\varepsilon_a n_a n_s \gamma'' \beta'' \sin^2 \theta \cos(\theta - \theta_t) \right\} \right]$$

Since the general expression for the ellipsometric coefficient ($r \equiv r_p/r_s$) in terms of the optical invariants is rather complicated, we only give the ellipsometric coefficient at an angle of incidence equal to the Brewster angle of the bare substrate for small values of the invariants

$$r(\theta_B^{\text{bare}}) = -i \frac{\omega}{c} \frac{\sqrt{\varepsilon_a + \varepsilon_a}}{\varepsilon_a - \varepsilon_s} I_c - \left(\frac{\omega}{2c} \right)^2 \frac{1}{\varepsilon_a - \varepsilon_s} (I_c)^2 - \left(\frac{\omega}{c} \right)^2 \frac{\varepsilon_a + \varepsilon_s}{\varepsilon_a - \varepsilon_s} I_t + \left(\frac{\omega}{c} \right)^2 \frac{\varepsilon_a \varepsilon_s}{\varepsilon_a - \varepsilon_s} I_{\delta e} \quad (2.32)$$

The ellipsometric angles follow in the usual way using $r \equiv \tan \psi \exp i\Delta$. The ellipsometric coefficient at the Brewster angle of the bare substrate gives the invariant I_e up to linear order. The reflectance for p-polarized light gives γ'' up to linear order and I_e up to quadratic order. As a consequence it is most convenient to obtain the real part of the invariant I_e by measuring the ellipsometric coefficient at the Brewster angle of the bare substrate.

Another quantity which is commonly measured is the shift of the Brewster angle owing to the film material. Defining the Brewster angle as the angle for which $\{Re\}(r(\theta_B)) = 0$, one finds for low values of the invariants

$$\theta_B = \theta_B^{\text{bare}} + \frac{n_a n_s \varepsilon_s}{(\varepsilon_s - \varepsilon_a)^2 (\varepsilon_s + \varepsilon - a)^3} \times \left[\frac{\omega}{c} (\varepsilon_s - \varepsilon_a) (\varepsilon_s + \varepsilon_a)^{3/2} I_c'' + \frac{1}{2} \left(\frac{\omega}{c} \right)^2 \times \left\{ \varepsilon_a (3\varepsilon_s - \varepsilon_a) (I_c'')^2 - (\varepsilon_s^2 - \varepsilon_a^2) (I_e'')^2 - 4(\varepsilon_s^2 - \varepsilon_a^2) [(\varepsilon_s + \varepsilon_a) I'_t - \varepsilon_a \varepsilon_s I'_{\delta e}] \right\} \right] \quad (2.33)$$

The refractive indices of the ambient and the substrate are $n_a = (\varepsilon_a)^{1/2}$ and $n_s = (\varepsilon_s)^{1/2}$. The angle of incidence and transmittance are θ and θ_t , respectively. The relation between these angles follow from Snellius law. Notice that since the ambient and the substrate are non-absorbing, the imaginary part of γ and β are also invariant, cf. Eqs. (2.19) and (2.20). These imaginary parts give the leading terms in the reflectance, transmittance and absorption, whereas the real and imaginary parts of I_e , and therefore of γ and β , both give leading terms of the same order to the ellipsometric coefficient. For light incident from the substrate, one only has to make the following changes $\varepsilon_a \leftrightarrow \varepsilon_s$ and $n_a \leftrightarrow n_s$ in the formulae of this section. The constitutive coefficients δ and τ change sign, since the z axis is now directed in the opposite direction, cf. Section 3.

3. Constitutive coefficients

3.1. Island films

In this section we give expressions for the constitutive coefficients for films consisting of particles on a plane substrate. The particles are embedded in the ambient. In order to calculate the constitutive coefficients, we will first pay attention to the response function of one particle. This response function [4] relates the polarization density in the particle to the incident electric field in the ambient substrate.

$$\vec{p}(\vec{x}_1, \omega) = \int d\vec{x}_2 \vec{A}(\vec{x}_1, \vec{x}_2, \omega) \cdot \vec{E}_a(\vec{x}_2, \omega) \quad (3.1)$$

At this point we want to emphasize that $\vec{A}$ is affected by the proximity of the substrate. The reason to go through the rather standard analysis below is to show that this dependence on the distance to the substrate has no effect on the validity of the formal expressions for the resulting polarizabilities. Expanding the potential [5] outside the sphere due to the polarization density inside the sphere,

$$\begin{aligned} \phi(\vec{x}_1) = \int d\vec{x}_2 \frac{\vec{p}(\vec{x}_2) \cdot (\vec{x}_1 - \vec{x}_2)}{|\vec{x}_1 - \vec{x}_2|^3} = & \left\{ \int d\vec{x}_2 \vec{p}(\vec{x}_2) \cdot \frac{\vec{x}_1}{|\vec{x}_1|^3} \right. \\ & + \left. \left\{ \int d\vec{x}_2 \vec{p}(\vec{x}_2) \vec{x}_2 \right\} \cdot \frac{(3\vec{x}_1 \vec{x}_1 - 1\vec{x}_1 \cdot \vec{x}_1)}{|\vec{x}_1|^5} \right\} \\ & + \text{higher order terms} \end{aligned} \quad (3.2)$$

one finds for the total polarization

$$\vec{p}_{\text{tot}} = \int d\vec{x}_2 \vec{p}(\vec{x}_2) \quad (3.3)$$

which is of course the sum of all the dipoles, and for the total quadrupole polarization

$$\vec{q}_{\text{tot}} = 3\vec{c} - 1\text{trace}(\vec{c}) \quad (3.4)$$

with

$$\vec{c} = \int d\vec{x} \{ \vec{p}(\vec{x})\vec{x} + \vec{x}\vec{p}(\vec{x}) \}$$

Substituting Eq. (3.1) and expanding the incident field to linear order around the center of the particle, one finds for the components of the dipole and quadrupole moments

$$\begin{aligned} p_i^{\text{tot}} = & \int d\vec{x}_1 \int d\vec{x}_2 A_{ik}(\vec{x}_1, \vec{x}_2) E_{a,k}(\vec{0}) \\ & + \int d\vec{x}_1 \int d\vec{x}_2 A_{ik}(\vec{x}_1, \vec{x}_2) x_{2,j} \partial_j E_{a,k}(\vec{0}) \\ = & \int d\vec{x}_1 \int d\vec{x}_2 A_{ik}(\vec{x}_1, \vec{x}_2) E_{a,k}(\vec{0}) \\ & + \int d\vec{x}_1 \int d\vec{x}_2 A_{ki}(\vec{x}_1, \vec{x}_2) x_{1,j} \partial_j E_{a,k}(\vec{0}) \end{aligned} \quad (3.5)$$

$$\begin{aligned} \vec{q}_{ij}^{\text{tot}} = & \int d\vec{x}_1 \int d\vec{x}_2 \{ 3A_{ik}(\vec{x}_1, \vec{x}_2) x_{1,j} - \delta_{ij} A_{ik}(\vec{x}_1, \vec{x}_2) x_{1,l} \\ & + i \leftrightarrow j \} E_{a,k}(\vec{0}) \end{aligned} \quad (3.6)$$

where one has to sum over double indices. The electric field has been expanded with respect to the origin, in such a way, that if one multiplies the total dipole and quadrupole polarization with the density of particles, we have an expansion up to second order in the particle size. Notice that we have chosen the origin of the coordinate frame in the center of the particle. Furthermore, we have used the fact, that the response function satisfies the following conditions:

$$A_{ik}(\vec{x}_1, \vec{x}_2) = A_{ki}^*(\vec{x}_2, \vec{x}_1) \quad (3.7)$$

$$A_{ik}(\vec{x}_1, \vec{x}_2) = A_{ik}^*(\vec{x}_1, \vec{x}_2) \quad (3.8)$$

The first relation is a consequence of microscopic time reversal invariance [4]. The second states that the response function must be real-valued.

One could argue that in the above calculation we have neglected effects due to retardation, which could conceivably be of the same order. However, the response function must satisfy

$$\vec{A}(\vec{x}_1, \vec{x}_2, \omega) = \vec{A}(\vec{x}_1, \vec{x}_2, -\omega) \quad (3.9)$$

As a consequence we have

$$\begin{aligned} \vec{A}(\vec{x}_1, \vec{x}_2, \omega) = & \vec{A}(\vec{x}_1, \vec{x}_2, 0) + \omega^2 \vec{A}_2(\vec{x}_1, \vec{x}_2) \\ & + \text{higher order terms} \end{aligned} \quad (3.10)$$

The first term on the right hand side is simply the static response function. The second term gives the first correction due to retardation effects. This term would, however, give terms of third and higher order in Eqs. (3.5) and (3.6).

Writing

$$p_i^{\text{tot}} = \alpha_{ij}^1 E_{a,j}(\vec{0}) + \alpha_{ijk}^{01} \partial_j E_k^a(\vec{0}) \quad (3.11)$$

$$q_{ij}^{\text{tot}} = \alpha_{ijk}^{10} E_{a,k}(\vec{0}) \quad (3.12)$$

we have for the polarizabilities with respect to the center of the particle

$$\begin{aligned} \alpha_{\perp,\perp}^1 = & \int d\vec{x}_1 \int d\vec{x}_2 A_{\perp,\perp}(\vec{x}_1, \vec{x}_2) \\ \alpha_{\parallel,\parallel}^1 = & \int d\vec{x}_1 \int d\vec{x}_2 A_{\parallel,\parallel}(\vec{x}_1, \vec{x}_2) \\ \alpha_{\parallel,\parallel,\parallel}^{10} = & 4\alpha_{\parallel,\parallel,\parallel}^{01} = \int d\vec{x}_1 \int d\vec{x}_2 4A_{\parallel,\parallel}(\vec{x}_1, \vec{x}_2) x_{1,\parallel} \\ \alpha_{\perp,\perp,\perp}^{10} = & 4\alpha_{\perp,\perp,\perp}^{01} = \int d\vec{x}_1 \int d\vec{x}_2 4A_{\perp,\perp}(\vec{x}_1, \vec{x}_2) x_{1,\perp} \\ \alpha_{\perp,\parallel,\perp}^{10} = & \alpha_{\perp,\perp,\perp}^{10} = 3\alpha_{\perp,\perp,\perp}^{01} = \int d\vec{x}_1 \int d\vec{x}_2 3A_{\perp,\perp}(\vec{x}_1, \vec{x}_2) \vec{x}_{1,\parallel} \\ \alpha_{\parallel,\perp,\parallel}^{10} = & \alpha_{\perp,\perp,\parallel}^{10} = 3\alpha_{\parallel,\perp,\parallel}^{01} = \int d\vec{x}_1 \int d\vec{x}_2 3A_{\parallel,\parallel}(\vec{x}_1, \vec{x}_2) x_{1,\perp} \end{aligned} \quad (3.13)$$

In the above expression, the particle and therefore $\vec{A}$ is invariant with respect to rotations around the axis normal to the ambient/substrate interface. This is done to simplify the expressions. The analysis of this section can easily be extended to arbitrary particles.

If we had chosen to develop the electric field with respect to the point $(0, 0, R)$, where R may be positive or negative, one finds for the polarizabilities with respect to the point $(0, 0, R)$, in terms of the polarizabilities given above,

$$\begin{aligned}\alpha_{\perp,\perp}^1(0, 0, R) &= \alpha_{\perp,\perp}^1 \\ \alpha_{\parallel,\parallel}^1(0, 0, R) &= \alpha_{\parallel,\parallel}^1 \\ \alpha_{\parallel,\parallel,\parallel}^{01}(0, 0, R) &= \alpha_{\parallel,\parallel,\parallel}^{01} \\ \alpha_{\perp,\perp,\perp}^{01}(0, 0, R) &= \alpha_{\perp,\perp,\perp}^{01} - R\alpha_{\perp,\perp}^1 \\ \alpha_{\perp,\parallel,\perp}^{01}(0, 0, R) &= \alpha_{\perp,\parallel,\perp}^{01} \\ \alpha_{\parallel,\perp,\parallel}^{01}(0, 0, R) &= \alpha_{\parallel,\perp,\parallel}^{01} - R\alpha_{\parallel,\parallel}^1\end{aligned}\quad (3.14)$$

If one chooses for R the (positive or negative) distance to the substrate and one chooses the X - Y plane parallel to the surface of the substrate, one may use the above expressions to place all the dipole and quadrupole moments on the surface of the substrate. Comparing the transformation properties of α_{ij}^1 and α_{ijk}^{01} with the expression for the constitutive coefficients in Eq. (2.18) in this equation ε_s should be replaced by ε_a since the multipoles are shifted in the medium with dielectric constant ε_a . One can then identify the constitutive coefficients for a film of identical particles with

$$\begin{aligned}\gamma &= \rho\alpha_{\parallel,\parallel}^1 \\ \beta &= \rho\alpha_{\perp,\perp}^1/\varepsilon_a^2 \\ \tau &= -\rho\alpha_{\parallel,\perp,\parallel}^{01} + \rho R\alpha_{\parallel,\parallel}^1 \\ \delta &= -\rho\alpha_{\parallel,\perp,\parallel}^{01}/\varepsilon_a + \rho R\alpha_{\parallel,\parallel}^1/\varepsilon_a - \rho\alpha_{\perp,\perp,\perp}^{01}/\varepsilon_a + \rho R\alpha_{\perp,\perp}^1/\varepsilon_a\end{aligned}\quad (3.15)$$

with ρ the number of particles per unit surface area and the multipole polarizabilities of the particles defined with respect to their center. Notice that the location of the dividing surface is chosen on the ambient/substrate interface. Since β is defined with respect to the normal displacement field, β is equal to $\rho\alpha_{\perp,\perp}^1/\varepsilon_a^2$ [2,3] instead of being equal to $\rho\alpha_{\perp,\perp}^1$. For light incident from the substrate, these coefficients must be taken with an extra minus sign and ε_a and ε_s must be interchanged in the expressions of Section 2.

In a previous article [1] we calculated the multipole polarizabilities of interacting spheres on a substrate. For the constitutive coefficients of films consisting of equally sized spheres with radius R , one finds using Eq. (3.15) and the expressions for the dipole and quadrupole polarizability for one sphere given in the general introduction, cf. Eqs. (1.1) and (1.2)

$$\gamma = 4\delta\varepsilon_a R \frac{\varepsilon - \varepsilon_a}{\varepsilon + 2\varepsilon_a} M_{11}^{11} \quad (3.16)$$

$$\beta = 4\phi \frac{R}{\varepsilon_a} \frac{\varepsilon - \varepsilon_a}{\varepsilon + 2\varepsilon_a} M_{10}^{10} \quad (3.17)$$

$$\tau = 4\phi\varepsilon_a R^2 \frac{\varepsilon - \varepsilon_a}{\varepsilon + 2\varepsilon_a} \left\{ M_{11}^{11} - \frac{\sqrt{5}}{3} M_{21}^{11} \right\} \quad (3.18)$$

$$\delta = 4\phi R^2 \frac{\varepsilon - \varepsilon_a}{\varepsilon + 2\varepsilon_a} \left\{ M_{11}^{11} - \frac{\sqrt{5}}{3} M_{21}^{11} + M_{10}^{10} - \frac{1}{2} \sqrt{\frac{5}{3}} M_{20}^{10} \right\} \quad (3.19)$$

The coverage ϕ is defined as $\rho\pi R^2$. It is important to notice that no approximations have been made in the derivation of these expressions for γ , β , δ and τ in terms of the M 's. An exact calculation of the M 's therefore leads to exact values of these constitutive coefficients without any adjustable parameters. In our first paper [1] we showed that such an exact calculation is possible for regular arrays and uncorrelated distributions of equally sized spheres. For random distributions of spheres we used a mean field approximation.

3.2. Stratified media

In order to put the theory given in Section 2 in more perspective, we will now also apply it to stratified media and show that it reproduces the results in the book of Lekner [1]. Consider a dielectric tensor profile $\varepsilon(\vec{z}) = \varepsilon_{\parallel}(z)(\vec{1} - \hat{z}\hat{z}) + \varepsilon_{\perp}(z)\hat{z}\hat{z}$ with $\varepsilon(\vec{z}) = \varepsilon_s \vec{1}$ for $z \rightarrow \infty$ and $\varepsilon(\vec{z}) = \varepsilon_a \vec{1}$ for $z \rightarrow -\infty$. A thin layer between z and $z + dz$ has constitutive coefficients given by

$$\begin{aligned}\gamma(z) &= [\varepsilon_{\parallel}(z) - \varepsilon^0(z)]dz \\ \beta(z) &= \left[\frac{1}{\varepsilon^0(z)} - \frac{1}{\varepsilon_{\perp}(z)} \right] dz \\ \tau(z) &= \delta(z) = 0\end{aligned}\quad (3.20)$$

The location of the dividing surface has been chosen at $z = 0$. The extrapolated bulk dielectric constant is then given by $\varepsilon^0(z) \equiv \varepsilon_s \theta(z) + \varepsilon_a \theta(-z)$. The Heaviside function $\theta(x - y)$ is defined by $\theta(x - y) = 1$ for $x < y$ and $\theta(x - y) = 0$ for $x \leq y$. Shifting this thin layer to $z = 0$ using Eq. (2.18) gives for the constitutive coefficients

$$\begin{aligned}\gamma(0; z) &= \gamma(z) \\ \beta(0; z) &= \beta(z) \\ \tau(0; z) &= -\gamma(z)z \\ \delta(0; z) &= -\gamma(z)z/\varepsilon^0(z) - \beta(z)z\varepsilon^0(z)\end{aligned}\quad (3.21)$$

In the argument the 0 gives the location of the shifted layer and the z indicates where this contribution originated from. In order to find the total effect of all these thin layers, one must apply the sum rule Eq. (2.21). After some calculations, this results in the following constitutive coefficients for the total layer:

$$\begin{aligned}
\gamma &= \int_{-\infty}^{\infty} dz [\varepsilon_{\parallel}(z) - \varepsilon^0(z)] \\
\beta &= \int_{-\infty}^{\infty} dz \left[\frac{1}{\varepsilon^0(z)} - \frac{1}{\varepsilon_{\perp}(z)} \right] \\
\tau &= - \int_{-\infty}^{\infty} dz [\varepsilon_{\parallel}(z) - \varepsilon^0(z)] z \\
\delta &= - \int_{-\infty}^{\infty} dz \left\{ \frac{1}{\varepsilon^0(z)} [\varepsilon_{\parallel}(z) - \varepsilon^0(z)] + \varepsilon^0(z) \right. \\
&\quad \times \left[\frac{1}{\varepsilon^0(z)} - \frac{1}{\varepsilon_{\perp}(z)} \right] \left. \right\} z - \frac{1}{2} \int_{-\infty}^{\infty} dz \int_{-\infty}^z dz' \\
&\quad \times \left\{ [\varepsilon_{\parallel}(z') - \varepsilon^0(z')] \left[\frac{1}{\varepsilon^0(z)} - \frac{1}{\varepsilon_{\perp}(z)} \right] \right. \\
&\quad \left. - [\varepsilon_{\parallel}(z) - \varepsilon^0(z)] \left[\frac{1}{\varepsilon^0(z')} - \frac{1}{\varepsilon_{\perp}(z')} \right] \right\} \quad (3.22)
\end{aligned}$$

The invariants follow upon substitution in Eqs. (2.19) and (2.20) and can be written after some calculations as

$$\begin{aligned}
I_e &= \int_{-\infty}^{\infty} dz \left[\varepsilon_{\parallel}(z) - \varepsilon_s - \varepsilon_a + \frac{\varepsilon_a \varepsilon_s}{\varepsilon_{\perp}(z)} \right] \\
I_{\tau} &= - \int_{-\infty}^{\infty} dz \int_{-\infty}^{\infty} dz' \\
&\quad \times \left\{ \frac{[\varepsilon_{\parallel}(z) - \varepsilon^0(z - z')][\varepsilon_{\parallel}(z') - \varepsilon^0(z' - z)]}{\varepsilon_s - \varepsilon_a} \right\} \\
I_{\delta e} &= \int_{-\infty}^{\infty} dz \int_{-\infty}^{\infty} dz' \\
&\quad \times \left\{ \frac{\varepsilon^0(z' - z)}{\varepsilon_s - \varepsilon_a} [\varepsilon_{\parallel}(z) - \varepsilon^0(z - z')] \right. \\
&\quad \times \left. \left[\frac{1}{\varepsilon_{\perp}(z')} - \frac{1}{\varepsilon^0(z' - z)} \right] \right\} \\
I_c &= \int_{-\infty}^{\infty} dz [(\varepsilon_a - \varepsilon_s)'' \varepsilon_{\parallel}(z) - (\varepsilon_a - \varepsilon_s)' \varepsilon_{\parallel}'(z) \\
&\quad - \varepsilon_a'' \varepsilon_s' + \varepsilon_a' \varepsilon_s''] \quad (3.23)
\end{aligned}$$

If one takes the parallel and normal component of the dielectric constant of the profile the same, the first three invariants are found to be equivalent to the integral invariants in Chapter 3.3 in Ref. [7]; the Eqs. (37), (31) and (39) correspond to $-I_e$, $(\varepsilon_s - \varepsilon_a)I_{\tau}$ and $(\varepsilon_s - \varepsilon_a)I_{\delta e}$, respectively. There is not an invariant equivalent to I_c in this reference. Expanding the denominators in the expressions of Section 2.2 and retaining only terms up to second order in the thickness of the layer, the resulting optical properties are equivalent to those in Chapter 12.5 in Ref. [7]. In this calculation, it was necessary to expand the denominators in the expressions of Section 2.2. One should notice, however, that in the theory presented here, both the denominators and nominators of the expressions in Section 2.2 are consistent up to second order in the film thickness. As

a consequence, it is not necessary to expand these denominators.

4. Numerical results for triangular arrays of gold spheres on a substrate

As an example we apply the theory of the previous sections to study the optical properties of small gold spheres situated in triangular arrays on a sapphire substrate. As in our previous article [1], the radius of the spheres is again chosen equal to 15 nm. For the dielectric constant, we took the values from Ref. [8] and included finite size corrections [9]. For the dielectric constant of sapphire, we took the values from Ref. [10].

There are some experimental results available for gold films with low coverage [11,12], which have been explained theoretically rather well using the constitutive coefficients β and γ for truncated spheres [13]. The non-spherical nature of the islands leads to significant differences with the predicted results for spheres.

In the following subsection, we shall first give the quadrupole approximation and study its range of validity. In the second subsection we discuss the results for the ellipsometric angles at the Brewster angle of the bare substrate and the change of the Brewster angle due to the film material.

4.1. The quadrupole approximation

According to the theory in the previous sections, the optical behaviour of thin films up to second order in the ratio of the film thickness to the wavelength is completely determined by the density of the static dipole and quadrupole polarizability in the film. It is therefore natural to introduce the so-called quadrupole approximation in which one only takes into account interactions between the particles and their images up to quadrupole order.

In Figs. 1 and 2 the relative percentage errors in the absolute values of the constitutive coefficients β and γ are presented for this quadrupole approximation in comparison with exact numerical results as a function of the coverage and the wavelength. The relative percentage errors in δ and τ have the same behaviour as the relative percentage error in γ . As already has been found in Ref. [1], the behaviour of β is less influenced by higher order multipoles than γ and the error is never more than 7%. Depending somewhat on the wavelength, we have for γ a reliability within about 1% or less for coverages ranging from 0 to about 70. For higher coverages, one finds differences up to almost 200%. This rather large discrepancy is related to the resonant behaviour of the parallel polarizability in this region. This resonant behaviour is denoted by the dashed line in Fig. 2. In our previous paper [1], we

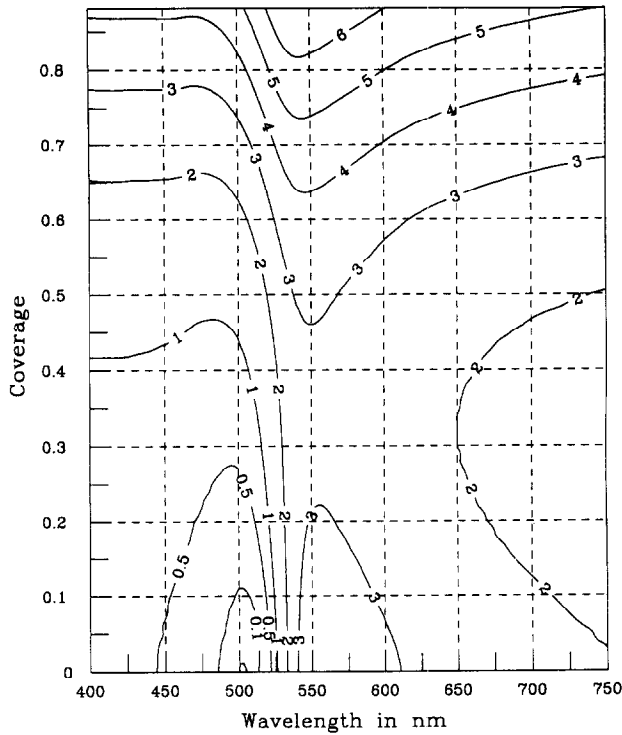


Fig. 1. The relative percentage errors in the absolute value of β in the quadrupole approximation as a function of the coverage and the wavelength.

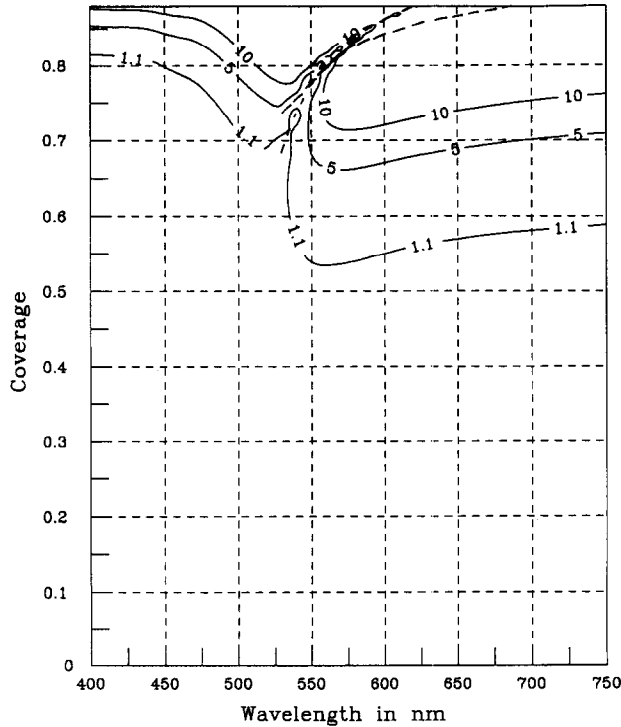


Fig. 2. The relative percentage errors in the absolute value of γ in the quadrupole approximation as a function of the coverage and the wavelength.

found that up to a coverage of 40% the optical properties for different sphere distributions, excluding fractal clustering, are practically the same and depend only on the coverage. In most cases an uncorrelated array of spheres on the substrate is appropriate. Here we have found that the so-called quadrupole approximation gives excellent results up to a coverage of about 50%. Therefore solving Eq. (30) of Ref. [1] up to quadrupole order for uncorrelated arrays (the solution is given below) is sufficient to describe the optical behaviour of these kind of films up to a coverage of about 40%. The error owing to neglecting the influence of the sphere distribution is never more than a few percent in this range. At coverages of about 50%, this error is about 5 to 10% depending somewhat on the optical property which one is using. In order to give the quadrupole approximation, we define the following coefficients

$$Q_{10}^{10} = 1 + \frac{1}{4}AB + \left\{ -\frac{1}{4}B\sqrt{2} + 1 \right\}A\phi$$

$$Q_{20}^{10} = -\frac{3}{80}BC\sqrt{15} + \frac{3}{160}\sqrt{30}C\phi$$

$$Q_{10}^{20} = -\frac{1}{16}AB\sqrt{15} + \frac{1}{32}\sqrt{30}A\phi$$

$$Q_{20}^{20} = 1 + \frac{3}{16}BC + \left\{ -\frac{9}{256}B\sqrt{2} + \frac{3}{16} \right\}C\phi$$

$$Q_{11}^{11} = 1 + \frac{1}{8}AB + \left\{ -\frac{1}{8}B\sqrt{2} - \frac{1}{2} \right\}A\phi$$

$$Q_{21}^{11} = -\frac{3}{80}BC\sqrt{5} + \frac{3}{160}\sqrt{10}C\phi$$

$$Q_{11}^{21} = -\frac{1}{16}AB\sqrt{5} + \frac{1}{32}\sqrt{10}A\phi$$

$$Q_{21}^{21} = 1 + \frac{1}{8}BC - \left\{ \frac{3}{128}B\sqrt{2} + \frac{1}{8} \right\}C\phi \quad (4.1)$$

with

$$A = \frac{\varepsilon - \varepsilon_a}{\varepsilon + 2\varepsilon_a}$$

$$B = \frac{\varepsilon_a - \varepsilon_s}{\varepsilon_a + \varepsilon_s}$$

$$C = \frac{2(\varepsilon - \varepsilon_a)}{2\varepsilon + 3\varepsilon_a} \quad (4.2)$$

The relevant matrix elements in the quadrupole approximation are then given by

$$M_{10}^{10} = Q_{20}^{20} / \{ Q_{10}^{10} Q_{20}^{20} - Q_{20}^{10} Q_{10}^{20} \}$$

$$M_{20}^{10} = -Q_{20}^{10} / \{ Q_{10}^{10} Q_{20}^{20} - Q_{20}^{10} Q_{10}^{20} \}$$

$$M_{11}^{11} = Q_{21}^{21} / \{ Q_{11}^{11} Q_{21}^{21} - Q_{21}^{11} Q_{11}^{21} \}$$

$$M_{21}^{11} = -Q_{21}^{11} / \{ Q_{11}^{11} Q_{21}^{21} - Q_{21}^{11} Q_{11}^{21} \} \quad (4.3)$$

The constitutive coefficients γ , β , τ and δ follow upon substitution in Eqs. (3.16)–(3.19).

4.2. Ellipsometric properties near the Brewster angle

In this section the ellipsometric angles at the Brewster angle of the bare substrate are studied. Furthermore the Brewster angle is calculated for the interface including the film material. The ellipsometric angles are defined by

$$r \equiv \frac{r_{\parallel}}{r_s} \equiv \tan \psi \exp(i\Delta) \quad (4.4)$$

and the Brewster angle is defined by the angle where the real part of r is 0:

$$\text{Re}(r(\theta_B)) = 0 \quad (4.5)$$

In Fig. 3 the Brewster angle is given for the film of gold spheres as a function of the wavelength and the coverage. For most coverages and wavelengths, the Brewster angle never deviates much more than 10% from its Fresnel value and is given in good approximation by the quadrupole approximation. In the region where γ shows a strong resonant behaviour, the increase of the Brewster angle is dramatic and it reaches a maximum value of about 85.7° at a coverage of 0.88 and a wavelength of about 708 nm. Here the quadrupole approximation is not correct.

In Figs. 4 and 5 the ellipsometric angles Δ and ψ are given at an angle of incidence which is equal to the Brewster angle of the bare substrate. From the behaviour of ψ one can conclude that it is not a monoto-

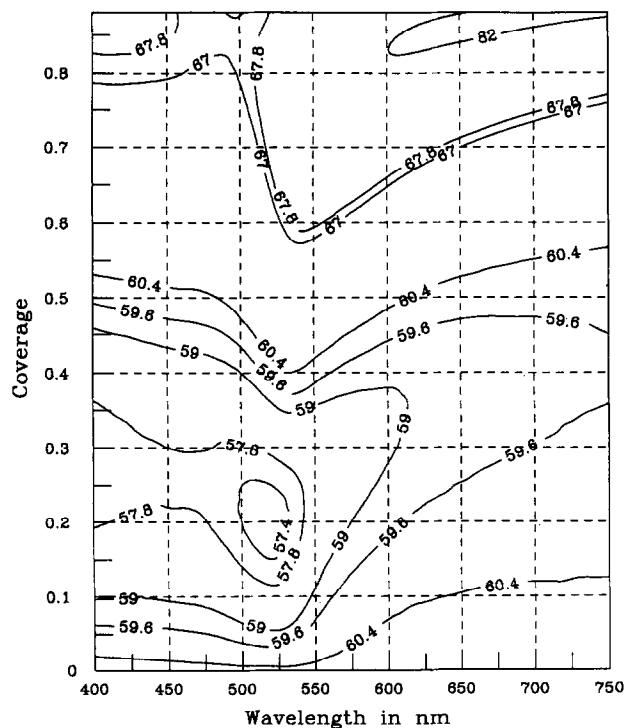


Fig. 3. The Brewster angle of the film material as a function of the coverage and the wavelength.

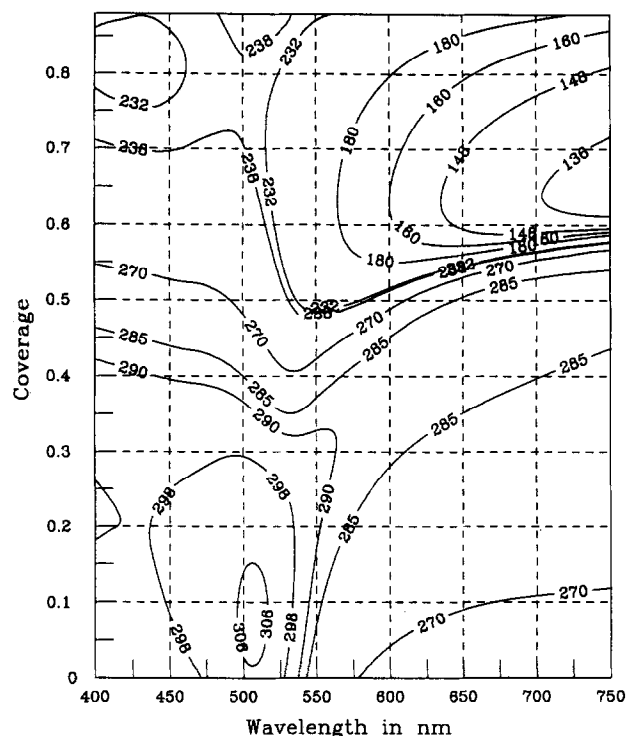


Fig. 4. The ellipsometric angle Δ at the Brewster angle of the bare substrate as a function of the coverage and the wavelength.

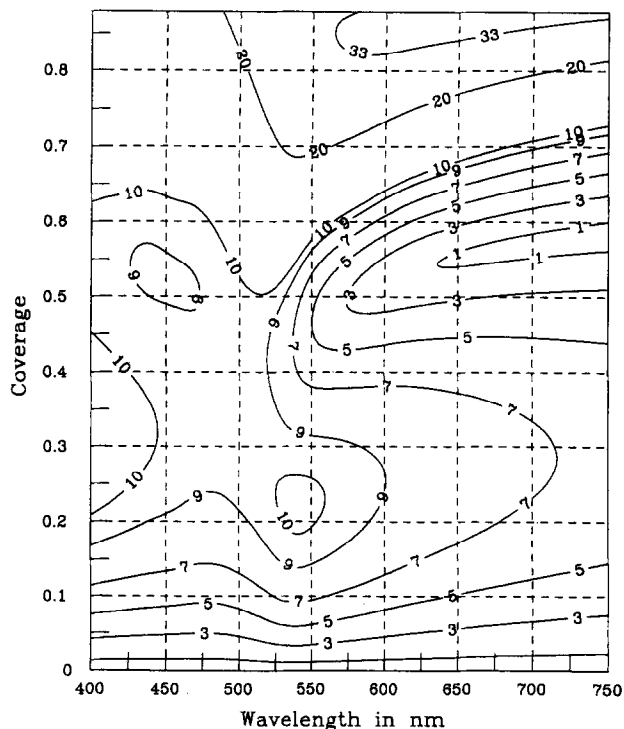


Fig. 5. The ellipsometric angle ψ at the Brewster angle of the bare substrate as a function of the coverage and the wavelength.

neously increasing function of the coverage except for coverages lower than 0.15. Up to a coverage of about 50% the quadrupole approximation works very well. Between a wavelength of about 625 nm and

This minimum occurs where Δ rapidly varies from a value of 290° to 180° in a range from about 0.05 around a coverage of 0.55. As a consequence Δ can be very close to 270° and ψ can be very close to zero for a coverage of about 0.55, this strongly resembles the ellipsometric behaviour at zero coverage. In the region where γ is resonant, ψ has a strong enhancement. Here Δ depends strongly on the wavelength but the resonant behaviour of the parallel polarizability is not apparent.

5. Summary

In this article we have given expressions for the optical properties of thin isotropic non-magnetic films for which the film structure can be described by a correlation length which is small in comparison with the wavelength of the incident light. These expressions are consistent up to second order in the ratio of the film thickness to the wavelength of the incident light and are given in terms of four universal constitutive coefficients β , γ , δ and τ , which completely characterize the optical behaviour of this kind of film. It is shown that the constitutive coefficients γ and β are related to the density of the static dipole polarizabilities in the film and that constitutive coefficients δ and τ are related to the density of the static quadrupole polarizabilities in the film. Retardation effects are of third and higher order. In order to find these coefficients, it is sufficient to solve the Laplace equation for the film in a homogeneous electric field. The dependence of γ , β , δ and τ on the choice of the so-called dividing surface is carefully studied. It is shown that the optical properties depend only on certain combinations of the constitutive coefficients which are independent of this choice and are called optical invariants. In experiments only these invariants are measured.

For island films on a plane substrate and for stratified media, we give explicit expressions for the constitutive coefficients. It is shown that for stratified media our theory gives the same results as found using the conventional methods [7]. In Ref. [6], measurements of R_p for a system of latex spheres in water absorbed on a glass substrate are described. These measurements are quite accurately predicted by the theory here presented using a simple quadrupole approximation for uncorrelated distributions of spheres to calculate the constitutive coefficients, without any adjustable parameters. In this article we have given the explicit expressions for the constitutive coefficients in this approximation, for uncorrelated distributions of spheres. Up to coverages of about 40%, these expressions may also be used for other sphere distributions.

The difference if one takes into account the precise nature of the sphere distributions is only a few percent in this range. For gold spheres on a sapphire substrate, we have studied the validity of this approximation by comparing with exact numerical results using the theory of a previous article [1]. It is found that up to coverages of about 50%, the quadrupole approximation gives quite accurate results. In order to illustrate the theory, we discuss the ellipsometric properties of gold spheres on a sapphire substrate in detail.

Owing to the the universal validity of the expressions for the optical properties for these kind of films, one can experimentally measure the values of the invariants without knowledge of the precise structure of the film by scanning some optical property as a function of the angle of incidence. If the model applied to calculate the invariants is realistic enough, the input parameters like film thickness, particle size, mean particle distance etc. may than be found from these invariants.

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